TA'S DETERMINACY PROBLEM FOR GOOD INVOLUTIONS OF CONJUGATION QUANDLES FAILS AT ORDER 32, AND, GRANTING THE PUBLISHED GROUP COUNTS, AT NO SMALLER ORDER

Abstract. For a finite group G let $\operatorname{Conj} G$ be G with the quandle operation $s_x(y) = xyx^{-1}$, let $k(G)$ be the class number, and let $\operatorname{Good}(\operatorname{Conj} G)$ be the set of good involutions of $\operatorname{Conj} G$ in the sense of Kamada. Problem 11.6 of Ta [4] asks whether $|\operatorname{Good}(\operatorname{Conj} G)|$ is determined by the triple $\big(n, |Z(G)|, k(G)\big)$, $n = |G|$. It is not. The dihedral group D of order 32 and the holomorph $\operatorname{Hol}(C_8)$ both have invariants $(32, 2, 11)$, yet

$$|\operatorname{Good}(\operatorname{Conj} D)| = 128, \qquad |\operatorname{Good}(\operatorname{Conj} \operatorname{Hol}(C_8))| = 1024.$$

Granting the published isomorphism-type counts of [1] for the orders $n \leq 31$ — external data that the accompanying program does not and cannot verify — the order 32 is the least at which this happens: for every nonabelian group of order at most 31 the triple does determine the count.

1. Introduction

Let G be a finite group. Its *conjugation quandle* $\operatorname{Conj} G$ is the set G equipped with $s_x(y) = xyx^{-1}$. Following Kamada [3], a *good involution* of a rack (X, s) is an involution $\rho \in \mathfrak{S}_X$ (that is, $\rho \circ \rho = \mathrm{id}_X$) such that

$$\rho \circ s_x = s_x \circ \rho \quad \text{and} \quad s_{\rho(x)} = s_x^{-1} \qquad \text{for all } x \in X. \tag{1}$$

Write $\operatorname{Good}(X, s)$ for the set of good involutions; the identity permutation belongs to it whenever it satisfies (1), and it is counted. Let $n = |G|$, let $Z(G)$ be the centre, let $\operatorname{Cl}(G)$ be the set of conjugacy classes and $k(G) = |\operatorname{Cl}(G)|$ the class number. For abelian G the quandle $\operatorname{Conj} G$ is trivial, so the interesting case is G nonabelian.

Ta [4, §11] closes an "Open questions" section, motivated by a computational census of the nonabelian groups of order at most 22, with the following.

Problem 11.6 of Ta. Let G be a finite nonabelian group of order n. Is the number of good involutions of $\operatorname{Conj} G$ completely determined by n, the order of the center $Z(G)$, and the class number $k(G)$ of G? If so, is there an exact formula to compute this number?

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Our answer to the first question is *no*. The refutation, Theorem 1.1, is unconditional; the accompanying statement that no smaller order refutes it, Proposition 1.2, is conditional on published isomorphism-type counts.

Theorem 1.1. *Let* $D = \langle r, s \mid r^{16} = s^2 = 1,\ srs = r^{-1}\rangle$ *be the dihedral group of order* 32 *and let*

$$H = \mathrm{Hol}(C_8) = C_8 \rtimes \mathrm{Aut}(C_8) = \langle a, b, c \mid a^8 = b^2 = c^2 = 1,\ bc = cb,\ bab^{-1} = a^{-1},\ cac^{-1} = a^5\rangle.$$

Both are nonabelian of order 32 *with* $|Z| = 2$ *and class number* 11*, and*

$$|\mathrm{Good}(\mathrm{Conj}\, D)| = 128, \qquad |\mathrm{Good}(\mathrm{Conj}\, H)| = 1024.$$

Consequently the answer to the first question of Problem 11.6 of [4] *is negative:* $|\mathrm{Good}(\mathrm{Conj}\, G)|$ *is not a function of* $(n, |Z(G)|, k(G))$.

Proposition 1.2. *Grant the isomorphism-type counts of* [1] *for each order* $n \le 31$*. If* G_1 *and* G_2 *are nonabelian groups of order at most* 31 *with* $|G_1| = |G_2|$*,* $|Z(G_1)| = |Z(G_2)|$ *and* $k(G_1) = k(G_2)$*, then*

$$|\mathrm{Good}(\mathrm{Conj}\, G_1)| = |\mathrm{Good}(\mathrm{Conj}\, G_2)|.$$

Only the first question is answered here; the second, which asks for an exact formula in those three invariants, is explicitly conditional on the first ("If so"), and with the antecedent false it is vacuous rather than false.

The characterisation of $\mathrm{Good}(\mathrm{Conj}\, G)$ on which both statements rest is not new: it is Lemma 2.1 below, proved in [4], and it already shows that the count is governed by the action of $Z(G)$ and of inversion on $\mathrm{Cl}(G)$ rather than by $k(G)$ alone. Given that lemma the two counts below are a short computation, and the census behind Proposition 1.2 is a short search. What remains is to exhibit a pair of groups on which those actions differ while the triple agrees, and — granting the published isomorphism-type counts — to check that no smaller order carries such a pair.

2. A reduction to labellings of the class set

The following lemma is the characterisation of the good involutions of $\mathrm{Conj}\, G$ proved in [4], in whose notation the labelling z appears as a central-valued class function. Its proof is included so that Theorem 1.1 depends on nothing quoted.

For $\zeta \in Z(G)$ and $C \in \mathrm{Cl}(G)$ the set $\zeta C^{-1} = \{\zeta x^{-1} : x \in C\}$ is again a conjugacy class, because ζ is central and inversion permutes classes; write $\iota_\zeta(C) = \zeta C^{-1}$. Each ι_ζ is an involution of $\mathrm{Cl}(G)$, since $\zeta(\zeta x^{-1})^{-1} = x$.

Lemma 2.1. *Let* G *be a finite group. The map* $z \mapsto \rho_z$*, where* $\rho_z(x) = z(C_x)\, x^{-1}$ *and* C_x *is the class of* x*, is a bijection from*

$$\Big\{\, z\colon \mathrm{Cl}(G) \to Z(G) \;\Big|\; z(\iota_{z(C)}(C)) = z(C) \text{ for all } C \in \mathrm{Cl}(G) \Big\}$$

onto $\mathrm{Good}(\mathrm{Conj}\, G)$*. In particular* $|\mathrm{Good}(\mathrm{Conj}\, G)|$ *depends only on the action of* $Z(G)$ *and of inversion on* $\mathrm{Cl}(G)$.

Proof. Let ρ be any permutation of G.

Step 1. For $x, y \in G$ one has $s_y = s_x^{-1}$ if and only if $ywy^{-1} = x^{-1}wx$ for all w, that is, if and only if xy centralises G. Hence $\{y : s_y = s_x^{-1}\} = Z(G)x^{-1}$, and the second condition in (1) says precisely that $\rho(x) = z(x)x^{-1}$ for a function $z\colon G \to Z(G)$.

Step 2. Write $w = xyx^{-1}$. With $\rho(u) = z(u)u^{-1}$ the first condition in (1) reads $z(w)w^{-1} = \rho(s_x(y)) = s_x(\rho(y)) = x\,z(y)y^{-1}x^{-1} = z(y)w^{-1}$, i.e. $z(xyx^{-1}) = z(y)$. So the first condition holds for all x, y if and only if z is constant on conjugacy classes, and we may regard z as a function on $\mathrm{Cl}(G)$.

Step 3. For such a z, $\rho(\rho(x)) = z(\rho(x))\rho(x)^{-1} = z(\rho(x))z(x)^{-1}x$, so ρ is an involution if and only if $z(\rho(x)) = z(x)$ for all x. The class of $\rho(x) = z(C_x)x^{-1}$ is $\iota_{z(C_x)}(C_x)$, which turns the last identity into $z\big(\iota_{z(C)}(C)\big) = z(C)$ for every class C. Distinct z give distinct ρ_z, and by Steps 1–3 every good involution arises this way. $\square$

Corollary 2.2. *Suppose $|Z(G)| = 2$, say $Z(G) = \{1, t\}$, and suppose every conjugacy class of G satisfies $C = C^{-1}$. Then $\mu(C) = tC$ is an involution of $\mathrm{Cl}(G)$ and*

$$|\mathrm{Good}(\mathrm{Conj}\, G)| = 2^{\,o},$$

where o is the number of orbits of μ. Equivalently, if μ has p two-cycles then $|\mathrm{Good}(\mathrm{Conj}\, G)| = 2^{\,k(G)-p}$.

Proof. Inverse-closure gives $\iota_1 = \mathrm{id}$ and $\iota_t = \mu$. In Lemma 2.1 the constraint at a class C with $z(C) = 1$ reads $z(C) = z(C)$ and is vacuous, while at a class with $z(C) = t$ it reads $z(\mu(C)) = t$. So the admissible z are exactly the indicator functions of the μ-invariant subsets $S = z^{-1}(t)$ of $\mathrm{Cl}(G)$, and since μ is an involution these are the unions of μ-orbits. There are 2^o of them, and $o = k(G) - p$ because the p two-cycles use $2p$ classes and the remaining $k(G) - 2p$ classes are fixed points. $\square$

Corollary 2.2 is used below only where both of its hypotheses hold; elsewhere the counts are obtained from Lemma 2.1 directly.

3. The two groups of order 32

The dihedral group of order 32. Let $D = \langle r, s \mid r^{16} = s^2 = 1,\ srs = r^{-1}\rangle$. Its centre is $Z(D) = \{1, r^8\}$, and its 11 conjugacy classes are

$$\begin{aligned}\mathrm{Cl}(D) = \{\ &\{1\},\ \{r^8\},\ \{r, r^{15}\},\ \{r^2, r^{14}\},\ \{r^3, r^{13}\},\ \{r^4, r^{12}\},\\ &\{r^5, r^{11}\},\ \{r^6, r^{10}\},\ \{r^7, r^9\},\\ &\{s, r^2s, r^4s, r^6s, r^8s, r^{10}s, r^{12}s, r^{14}s\},\\ &\{rs, r^3s, r^5s, r^7s, r^9s, r^{11}s, r^{13}s, r^{15}s\}\ \}.\end{aligned} \tag{2}$$

Every listed class equals its own inverse. Multiplication by $t = r^8$ sends the class of r^i to the class of $r^{i+8} = r^{-(8-i)}$, so it interchanges

$$\{1\} \leftrightarrow \{r^8\}, \qquad \{r\}^{\pm} \leftrightarrow \{r^7\}^{\pm}, \qquad \{r^2\}^{\pm} \leftrightarrow \{r^6\}^{\pm}, \qquad \{r^3\}^{\pm} \leftrightarrow \{r^5\}^{\pm},$$

writing $\{r^i\}^{\pm} = \{r^i, r^{-i}\}$, and it fixes $\{r^4, r^{12}\}$ as well as both reflection classes, since r^8 preserves the parity of the exponent. Thus μ has $p = 4$ two-cycles and 3 fixed points, $3 + 2 \cdot 4 = 11 = k(D)$, and Corollary 2.2 gives

$$|\mathrm{Good}(\mathrm{Conj}\, D)| = 2^{\,11-4} = 2^7 = 128.$$

The holomorph of C_8. Let $H = \mathrm{Hol}(C_8) = C_8 \rtimes \mathrm{Aut}(C_8)$, presented as $\langle a, b, c \mid a^8 = b^2 = c^2 = 1,\ bc = cb,\ bab^{-1} = a^{-1},\ cac^{-1} = a^5\rangle$, of order $8 \cdot 4 = 32$. Conjugation by $b^e c^f$ acts on $\langle a\rangle$ as $a \mapsto a^u$ with $u = 7^e 5^f \bmod 8$, and $\{7^e 5^f\} = \{1,3,5,7\} = \mathrm{Aut}(C_8) \cong C_2 \times C_2$. Hence $Z(H) = \{1, a^4\}$: an element a^i is central exactly when $2i \equiv 0 \pmod 8$, and no element outside $\langle a\rangle$ commutes with a. The classes are computed as follows. Inside $\langle a\rangle$ they are the $\mathrm{Aut}(C_8)$-orbits on $\mathbb{Z}/8$. In the coset $\langle a\rangle b$, conjugation by a^j sends $a^i b$ to $a^{i+2j} b$ and conjugation by b or c preserves the parity of i; in $\langle a\rangle bc$ the same computation with $u(1,1) = 3$ gives $a^i bc \mapsto a^{i-2j} bc$. In $\langle a\rangle c$ conjugation by a^j sends $a^i c$ to $a^{i-4j} c$ while conjugation by b sends it to $a^{-i} c$, which merges $\{ac, a^5 c\}$ with $\{a^3 c, a^7 c\}$. The outcome is the list

$$\begin{aligned} \mathrm{Cl}(H) = \{\ & \{1\},\ \{a^4\},\ \{a^2, a^6\},\ \{a, a^3, a^5, a^7\}, \\ & \{c, a^4 c\},\ \{a^2 c, a^6 c\},\ \{ac, a^3 c, a^5 c, a^7 c\}, \\ & \{b, a^2 b, a^4 b, a^6 b\},\ \{ab, a^3 b, a^5 b, a^7 b\}, \\ & \{bc, a^2 bc, a^4 bc, a^6 bc\},\ \{abc, a^3 bc, a^5 bc, a^7 bc\}\ \}, \end{aligned} \tag{3}$$

of size $4+3+2+2 = 11$, with class sizes $1+1+2+4+2+2+4+4+4+4+4 = 32$. Every class equals its own inverse. In $\langle a\rangle$ this is because $\mathrm{Aut}(C_8)$ contains -1. In $\langle a\rangle b$ every element is an involution, since $(a^i b)^{-1} = b a^{-i} = a^i b$. In $\langle a\rangle bc$ one has $(a^i bc)^{-1} = bc\, a^{-i} = a^{-3i} bc$, and $-3i$ has the parity of i, so inversion preserves each of the two classes there. In $\langle a\rangle c$ one has $(a^i c)^{-1} = c a^{-i} = a^{-5i} c$, which for odd i stays in the four-element class and for $i = 0, 2, 4, 6$ gives $0, 6, 4, 2$, again within the listed classes. Multiplication by $t = a^4$ interchanges $\{1\}$ and $\{a^4\}$ and fixes the other nine classes, since a^4 preserves each of the nine listed sets displayed above. Thus $p = 1$ and $9 + 2 \cdot 1 = 11 = k(H)$, so

$$|\mathrm{Good}(\mathrm{Conj}\, H)| = 2^{\,11-1} = 2^{10} = 1024.$$

Proof of Theorem 1.1. Both groups are nonabelian of order 32 with $|Z| = 2$ and $k = 11$, and their good-involution counts are 128 and 1024 by the two computations above. Since the counts differ, $D \not\cong H$, and $(n, |Z(G)|, k(G)) = (32, 2, 11)$ therefore does not determine $|\mathrm{Good}(\mathrm{Conj}\, G)|$. $\square$

4. Minimality, granting the published counts

Proof of Proposition 1.2. An order n can contribute a pair only if it carries at least two nonabelian isomorphism types. The number of groups of order n is the OEIS sequence `A000001` [1] and the number of abelian ones is $\prod_p p(a_p)$

for $n = \prod_p p^{a_p}$, where $p(\cdot)$ is the partition function; subtracting, the orders $n \le 31$ with two or more nonabelian types are exactly

$$8,\ 12,\ 16,\ 18,\ 20,\ 24,\ 27,\ 28,\ 30,$$

with $2, 3, 9, 3, 3, 12, 2, 2, 3$ types respectively. Every other $n \le 31$ has at most one nonabelian type — in particular $23, 29, 31$ are prime and $9, 25$ are squares of primes, with none at all — and there is nothing to compare. For each of the nine remaining orders the accompanying program constructs that many pairwise non-isomorphic nonabelian groups — the constructions are given in the program and are not printed here, and it is the program that computes the isomorphism invariants separating them — which meets the count taken from [1] and so, granting that count, completes the list; it then computes $(|Z|, k)$ and $|\mathrm{Good}(\mathrm{Conj}\, G)|$ for each by Lemma 2.1, and collects the groups into buckets of equal invariants. The buckets holding two or more groups are

$(8,2,5):$ $\{D_4, Q_8\}$ at 16,	$(20,2,8):$ $\{D_{10}, \mathrm{Dic}_5\}$ at 16,
$(12,2,6):$ $\{D_6, \mathrm{Dic}_3\}$ at 8,	$(24,2,9):$ 3 groups at 64,
$(16,2,7):$ 3 groups at 32,	$(24,4,12):$ 4 groups at 1000,
$(16,4,10):$ 6 groups at 2160,	$(24,6,15):$ $\{C_3{\times}D_4, C_3{\times}Q_8\}$ at 608000,
$(18,1,6):$ 2 groups at 1,	$(27,3,11):$ 2 groups at 324,
	$(28,2,10):$ $\{D_{14}, \mathrm{Dic}_7\}$ at 32.

(Here D_m is dihedral of order $2m$ and Dic_m dicyclic of order $4m$. At order 30 the three nonabelian groups D_{15}, $C_3 \times D_5$, $C_5 \times S_3$ have pairwise distinct invariants $(30,1,9)$, $(30,3,12)$, $(30,5,15)$, so no bucket is live.) Each live bucket is constant, which is the assertion. □

The completeness of each order's list is the one point at which the census rests on external data: that nine orders and no others are live, and that the constructed groups exhaust each of them, follows only from the published counts of [1], which are quoted and not verified. Everything else about those groups, including their pairwise non-isomorphism, is established inside the program; but a reader who declines to grant [1] is left with Theorem 1.1 alone. The counts computed for the nonabelian groups of order at most 22 agree with the table in Appendix B of [4], and their per-order sums with the published terms of [2].

Proposition 1.2 concerns orders at most 31; nothing here says that 32 is the only order at which the triple fails.

5. Verification

The computations above were re-run independently, in exact integer arithmetic. For the two groups of order 32 the only inputs are the presentations and the class lists (2) and (3) printed in §3; the census of Proposition 1.2 additionally consumes the isomorphism-type counts of [1] and constructions of the census groups that are not printed here. The program rebuilds each group as a Cayley table and checks associativity on all n^3 triples; verifies the three set-level equivalences of the proof of Lemma 2.1 by exhaustion over all $x, y \in G$ and all pairs of central values; compares the class lists (2) and (3) element by element with the classes it computes; counts $\mathrm{Good}(\mathrm{Conj}\, G)$ by a matching recursion over $\mathrm{Cl}(G)$ and by a constraint enumeration, and for the two groups of order 32 also by Corollary 2.2 and by a fourth route that tests *every* function $\mathrm{Cl}(G) \to Z(G)$ against (1) literally; and re-runs the census of Proposition 1.2. For S_3, D_4, Q_8 and D_5 it also scans every self-inverse permutation of G, recovering 1, 16, 16, 1; this pins the convention, in force throughout, that the identity permutation is counted, under which Appendix B of [4] likewise reports 16 for $\mathrm{Conj}\, D_4$. Every check passed. For each of the nine live orders the recorded run prints the number of pairwise non-isomorphic nonabelian groups exhibited beside the quoted term of [1] and the abelian count subtracted from it, so that the one external input to Proposition 1.2 can be read off and checked against the source; it also prints every census group with its $\big(|Z|, k, |\mathrm{Good}(\mathrm{Conj}\, G)|\big)$. Granting those quoted terms, the run settles every order $6 \le n \le 31$; it does not establish them. The program and its recorded run accompany this note, and the run states its own scope, including that [1] is external input; it also records checks on material not discussed here, on which nothing above depends.

References

[1] OEIS Foundation Inc., *Sequence A000001: number of groups of order n*, The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A000001.

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